

What is AWS VPC (Virtual Private Cloud)

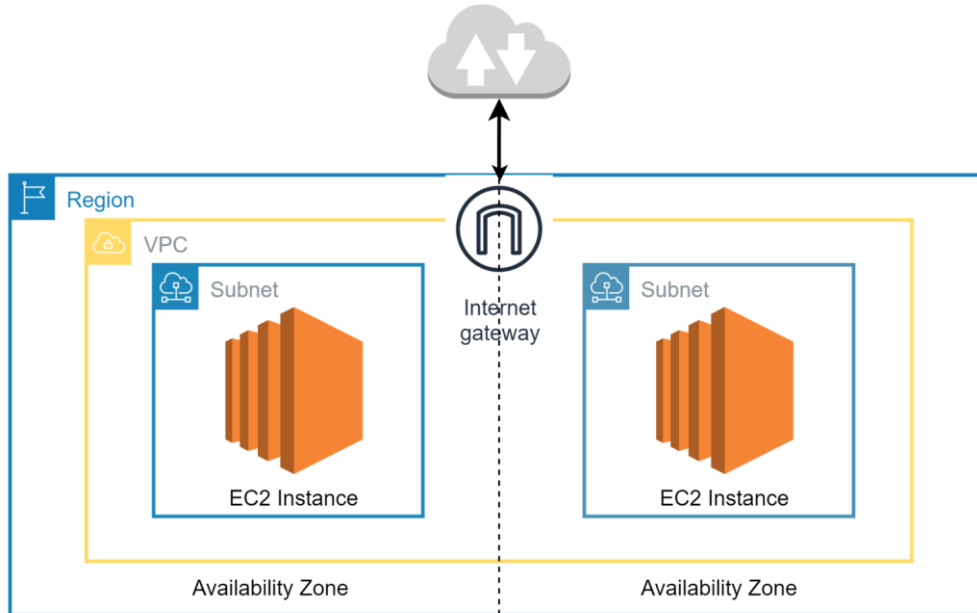
Basic Network understanding

Part1

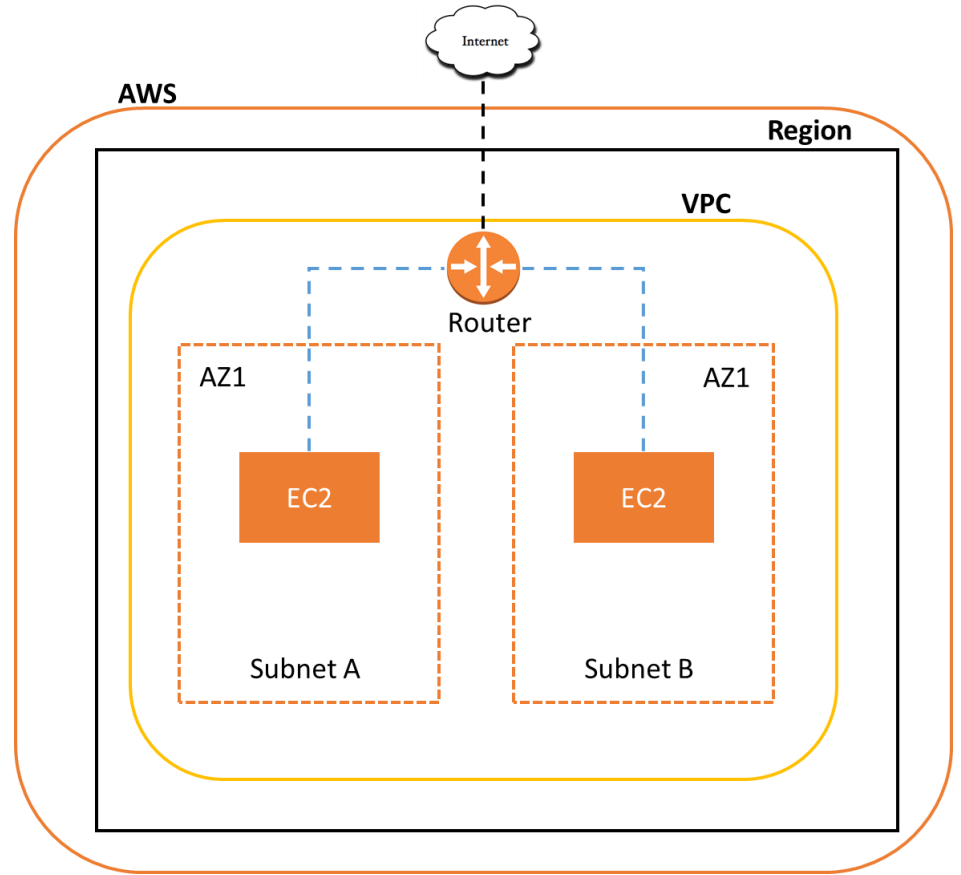
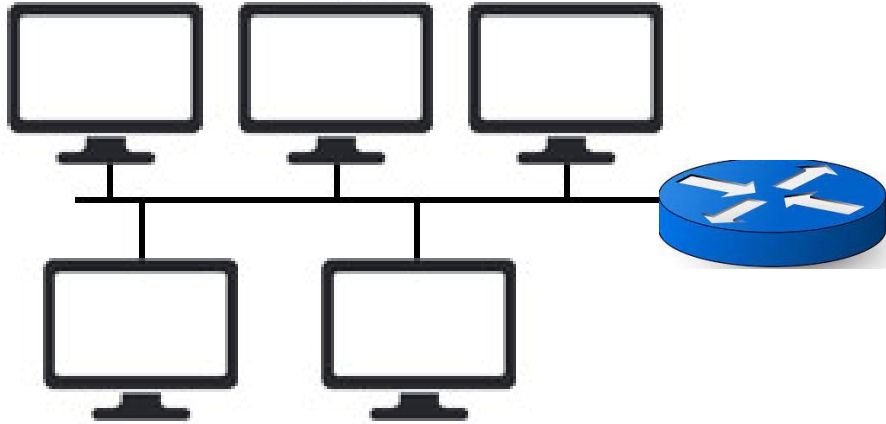
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What is Amazon VPC?

Amazon Virtual Private Cloud (Amazon VPC) enables you to launch AWS resources into a virtual network that you've defined. This virtual network closely resembles a traditional network that you'd operate in your own data center, with the benefits of using the scalable infrastructure of AWS.



Traditional Hosting Vs VPC



Features

The following features help you configure a VPC to provide the connectivity that your applications need:

Virtual private clouds

A [VPC](#) is a virtual network that closely resembles a traditional network that you'd operate in your own data center. After you create a VPC, you can add subnets.

Subnets

A [subnet](#) is a range of IP addresses in your VPC. A subnet must reside in a single Availability Zone. After you add subnets, you can deploy AWS resources in your VPC.

IP addressing (IPv4/IPv6)

You can assign [IP addresses](#), both IPv4 and IPv6, to your VPCs and subnets. You can also bring your public IPv4 and IPv6 GUA addresses to AWS and allocate them to resources in your VPC, such as EC2 instances, NAT gateways, and Network Load Balancers.

Features

Routing



Use [route tables](#) to determine where network traffic from your subnet or gateway is directed.

Gateways and endpoints



A [gateway](#) connects your VPC to another network. For example, use an [internet gateway](#) to connect your VPC to the internet. Use a [VPC endpoint](#) to connect to AWS services privately, without the use of an internet gateway or NAT device.

Peering connections



Use a [VPC peering connection](#) to route traffic between the resources in two VPCs.

Traffic Mirroring



[Copy network traffic](#) from network interfaces and send it to security and monitoring appliances for deep packet inspection.

Features

Transit gateways



Use a [transit gateway](#), which acts as a central hub, to route traffic between your VPCs, VPN connections, and AWS Direct Connect connections.

VPC Flow Logs



A [flow log](#) captures information about the IP traffic going to and from network interfaces in your VPC.

VPN connections



Connect your VPCs to your on-premises networks using [AWS Virtual Private Network \(AWS VPN\)](#).

VPC Concept

VPC

Security Group - It controls the inbound and outbound traffic for the instance.

NACL- network access control list is acts as a firewall for controlling traffic in and out of one or more subnets

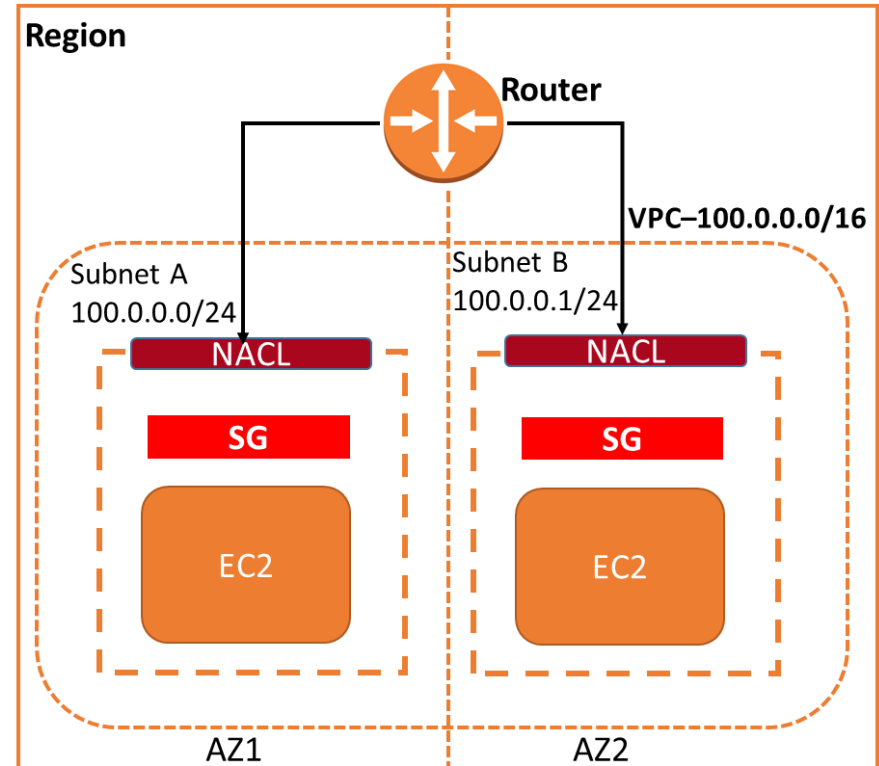
Route Table- It shows local route

Subnet - It is a range of IP addresses in your VPC

Availability Zone(AZ)- highly available data centers within each AWS region

Region - physical locations around the world

CIDR- (Classless Inter-Domain Routing) is a method of assigning IP addresses
100.0.0.0/16



Difference between Security Groups and Network ACLs

Security Group	NACL
Firewall of EC2 Instances	Firewall of the Subnet
Stateful - changes applied to an incoming rule will be automatically applied to the outgoing rule	Stateless - any changes applied to an incoming rule will not be applied to the outgoing rule
support allow rules only Example: You cannot deny a certain IP address from establishing a connection	support allow and deny rules Example: you could explicitly deny a certain IP address to establish a connection

VPC Features

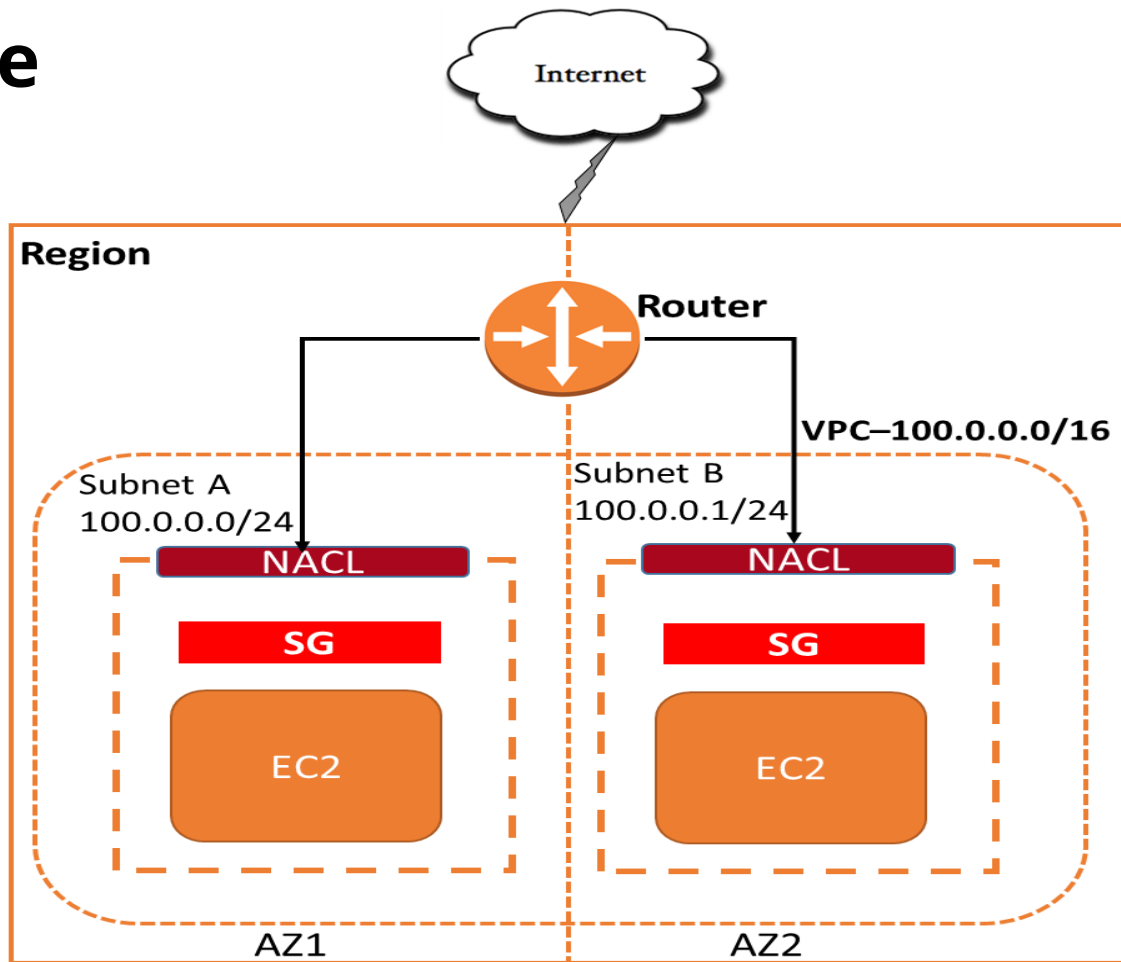
Subnet:

A subnet is a range of IP addresses in your VPC. You can launch AWS resources, such as EC2 instances, into a specific subnet. When you create a subnet, you specify the IPv4 CIDR block for the subnet, which is a subset of the VPC CIDR block.

Subnet Types:

- ❖ **Public Subnet:** The subnet traffic is routed to the public internet through an internet gateway or an egress-only internet gateway.
- ❖ **Private Subnet:** The subnet traffic can't reach the public internet through an internet gateway or egress-only internet gateway. Access to the public internet requires a NAT device.
- ❖ **VPN-only subnet :** The subnet traffic is routed to a Site-to-Site VPN connection through a virtual private gateway. The subnet traffic can't reach the public internet through an internet gateway.

Exercise



Thank You